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March 15, 2025

To Whom It May Concern:

This document responds to the Request for Information (RFI) issued by the NITRD NCO on behalf of the OSTP for input from all interested parties on the Development of an Artificial Intelligence (AI) Action Plan ("Plan") to identify the highest priority policy actions that should be in the new AI Action Plan. This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.

This individual response[1] addresses two of the actions on the list and adds a third that suggests the Plan include an approach to develop trusted AI data.

1. Technical and Safety standards

Simply put, standards should not be seen as "an end unto themselves". The government is not exempt from the management mantra that what you measure is what you get. Rather than simply calling for standards to be published, which will lead to a proliferation of standards, the plan should call for standards that are adopted and that are evaluated to assess whether the standards are actually meeting the stated goals.

A recent National Institute of Standards and Technology (NIST) Visiting Committee on Advanced Technology report stated "the committee's view is that we currently lack frameworks for measuring the aggregate impacts of standardization, along with the tools and metrics required for assessing the effect of policy interventions in this domain." [2] The report noted that this is at least partly due to the "lack of any formal or shared method to measure investments in standardization or their impacts". That Committee recommended that "NIST establish a project in collaboration with academia and the standards community to create a defined set of objectives, conceptual framework, taxonomy of metrics, and

[1] These views are personal and do not represent any institution with which I am or have been affiliated. The following list discloses those institutions to ensure full transparency. I served on the National AI Research Resources (NAIRR) Taskforce and currently serve on the Advisory Committee for the NAIRR Pilot. I have recently completed a one year IPA at the National Institute for Standards and Technology. I have been deeply involved with the National Science Foundations Technology, Innovation and Partnership program's workforce work. I serve on the Advisory Committee for the National Data Platform.
[2] Report to NIST Leadership for the Implementation of the U.S. Standards Strategy for Critical and Emerging Technology by the Visiting Committee on Advanced Technology of the National Institute of Standards and Technology February 2024 https://www.nist.gov/director/vcat/reports

common qualitative factors for measuring both the value of investment in standards and their impact." (Recommendation 26)[1].

Recommendation: The Plan could charge an agency with establishing an AI standards evaluation framework that could be used to define and scope the definition of the intended outcomes associated with the development of AI standards. The Plan should charge the agency to work with the community to identify the components of the theory of change and associated measurement so that the framework could also be used to provide in federal, state, and local governments with early indications of standards effectiveness for monitoring purposes. An exemplar is provided below[3]

Illustrative Non-Exhaustive Examples

Inputs	Activities	Outputs	Outcomes	Goals
Establishment of input from Working Groups	Identify foundational concepts and technologies	Publish common terminologies and taxonomies	Adoption of and iteration with key actors about common terminologies that speed innovation and lower costs	Greater innovation; lower costs
Organization of and inputs from Workshops	Set norms for governance and accountability processes	Publish measurement methods and metrics	Adoption of and iteration with key actors about measurement methods and metrics that will inform understanding	More informed understanding of benefits and harm of AI
Research Results	Develop standards for training and evaluation datasets	Publish standards for training data processes	Adoption of and iteration with key actors about standards and improved understanding of bias and harm	Informed investments in AI standards based on what works
Organization of and input from International Engagement	Address interactions between AI systems and people and institutions	Publish standards for training data processes	Adoption of and iteration with key actors on transparency methods leading to broader public understanding of AI	Trustworthy AI system

Figure 3: Theory of Change for AI Standards

The Plan could recommend that once a theory of change is established for particular AI standards – or, indeed, for an entire class of products or processes such as the entity resolution use case – the framework could be used as a basis for providing information about progress at each step in a theory of change. That information could be used, in concert with stakeholders, to monitor progress, identify problems, and provide accountability and

[3] Source: Figure 3: NIST AI 100-7, an initial public draft entitled "Towards an Approach for Evaluating the Impact of AI Standards" authored by Julia Lane. The draft has been approved for, but is waiting, publication.

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transparency to the public. The theory of change could be adjusted and modified as the impacts are evaluated as described below[4].

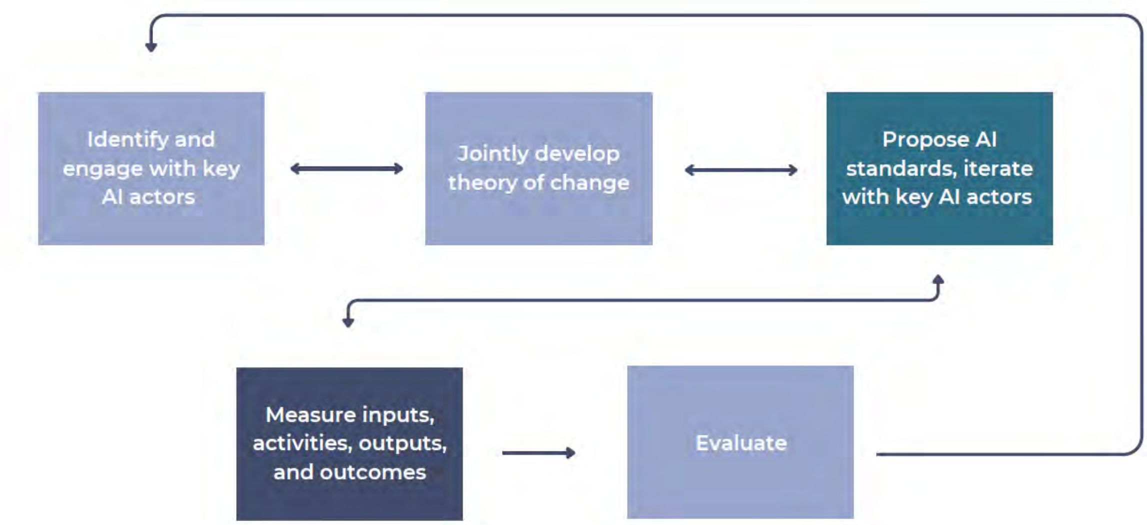

Figure 4: Engagement of key AI actors

The Plan should intentionally advertise both the establishment of an evaluation methodology and the development of a clear analytic framework that can be used to report the results of AI systems in order to build public trust.

2. Education, workforce, innovation and competition

The Plan should leverage two existing efforts to inform future investments in AI: the National AI Research Resources (NAIRR) pilot and the investments made by NSF's Technology, Innovation, and Partnerships (TIP) Directorate.

2a. NAIRR example: Much initial thinking was done by the NAIRR Task Force, which developed a detailed theory of change about the impacts of AI investment on these goals[5]. The NAIRR pilot has implemented many of the recommendations of the Taskforce, and has learned many lessons about implementation, measurement, and uptake[6]

[4] Source: Figure 4: NIST AI 100-7, an initial public draft entitled "Towards an Approach for Evaluating the Impact of AI Standards" which is approved for, but waiting publications.
[5] https://www.ai.gov/wp-content/uploads/2023/01/NAIRR-TF-Final-Report-2023.pdf
[6] https://nairrpilot.org/

Recommendations: The Plan should incorporate the Theory of Change in the NAIRR Task force report. The Plan should reference and build on NAIRR pilot experience when making additional investments. The Plan could consider further investments in the NAIRR which has already two years of experience and has excellent leadership.

2b. NSF TIP example The National Science Foundation's TIP Directorate has made great strides in understanding the impact of AI research investments in education, workforce, innovation and competition in funding the "Industries of Ideas" project[7]. That has been made possible by about 15 years of investments in the STAR METRICS/UMETRICS program, initiated by the NSTC and OSTP (in cooperation with NITRD)[8].

The Industries of Ideas project represents a fundamentally new measurement approach to measuring impact that could inform the Plan. Because AI is a swiftly changing technology, the measures need to be timely. Because the impacts are often very local, the measures need to be developed and validated by local stakeholders. Because the impacts are on both jobs and earnings at the firm and individual level and, the measures need to be developed using longitudinal linked employer-employee workforce data. Because the very use of the term "impact" requires a counterfactual, the measures need to be developed for both AI "touched" and AI "untouched" firms so the differences at the worker and firm level can be compared. And finally, for the results to be useful and informative for decision making, the measurement framework needs to be understandable. The measurement framework in the "industry of ideas" meets all of these requirements, and has strong support with a number of state workforce development agencies, labor market information agencies, economic development entities and training providers. The project could be extended to integrate skill demand data with dynamic information on real earnings and employment outcomes for the universe of workers in one state in a manner that is explicitly designed with input from a dozen engaged and committed other states to scale to their needs. The approach would provide better timely, local, and actionable information to all stakeholders. The results would align training, hiring, and workforce policies with the actual realities of the job market, transforming generic workforce development into targeted investments with measurable, evidence-based returns.

[7] https://iris.isr.umich.edu/tip Lane, Julia, Jason Owen-Smith, and Bruce A. Weinberg. "How to track the economic impact of public investments in AI." *Nature* [redacted] (2024): 302-304. https://www.nature.com/articles/d41586-024-01721-1 https://www.aei.org/research-products/report/the-industry-of-ideas-measuring-how-artificial-intelligence-changes-labor-markets/

[8] Lane, Julia, and Stefano Bertuzzi. "The STAR METRICS project: current and future uses for S&E workforce data." Science of Science Measurement Workshop, held Washington DC. Vol. 12. 2010.2. Valdez, B. and J. Lane, *The science of science policy: a federal research roadmap.* Report to The Subcommittee on Social, Behavioral and Economics Sciences, Committee on Science, National Science and Technology Council. Washington, DC: Office of Science and Technology Policy, Executive Office of the President, 2008.

The core approach is grounded in the notion that measuring the economy-wide impact of federal investments in AI R&D requires identifying the people at the heart of these investments. It is people – not documents - who generate ideas, launch start-ups and influence the next generation of innovators through academic and professional networks. In emerging industries, where ideas matter a lot, people are the main value-creating unit — not machines or office floor space. Tracking the impact of research investments requires capturing HR and finance information on all the people funded on research grants – students and doctoral or postdoctoral trainees as well as research staff and faculty collaborators. Many of these individuals might never publish a paper, file a patent or become a PI themselves. But conducting research teaches them about cutting-edge algorithms and the application of these technologies. The movement of these trainees and staff through to the wider economy, and the transmission of their ideas, is captured when they get jobs in the private sector. Their earnings and employment are recorded in state administrative data. This linkage — between academia and private-sector employment — is the new data layer that is being analyzed in a pilot project in Ohio that is likely to be extended to other states. The initial results? People employed: More than 470,000; Students and trainees employed: More than 290,000; Average number employed by a project lead: 22. Additional work is being done on the impact on business survival and growth.

Recommendations: The Plan could recommend that science agencies build on and expand the NSF/TIP investments in the data infrastructure. The Plan could recommend that the infrastructure be used by federal agencies to provide consistent and verifiable descriptive and causal estimates of the impact of research investments on the areas of interest.

3. Producing Trusted AI Data

Rigorous scientific discovery depends on high-quality data and reproducible empirical research, particularly in AI. The lack of high quality AI data is "holding AI science back"[9] - for example, the Protein Data Bank was critical to the discoveries that led to the 2024 Nobel Prize in Chemistry. Poor data and the resultant misinformation – particularly with the increased importance of data for artificial intelligence applications - can lead to harmful outcomes for both science and the public.

The problem to be addressed is how to signal what data can be trusted, and for what purpose. That requires information both on how data are produced and how data can be used. One approach that has been used in many other contexts is to develop community-driven data quality standards. There are at least three steps.

[9] https://www.technologyreview.com/2024/10/15/1105533/a-data-bottleneck-is-holding-ai-science-back-says-new-nobel-winner/

Step 1 is to identify a community of data producers and users using a specific dataset or set of datasets to inform their AI research that .

Currently, identifying a research community by data use is challenging. There are only relatively ad-hoc methods to find what different datasets measure, what research has been done using those data by which researchers, with what code, and with what results. The resulting lack of information leads to an underestimating of the value of investing in data which in turn leads both agencies and researchers to underinvest in creating, documenting, and disseminating datasets. Since agencies cannot determine the return on investment in data, they cannot appropriately allocate resources. And since researchers who create and improve datasets receive limited rewards for their subsequent and unknown reuse, their incentives to invest in and disseminate research data are dulled – many researchers who say they will share their data don't[10].

Recommendation: Develop automated tools to identify communities of data users[11] as the first step to rewarding agency and researcher contributions to data documentation and dissemination[12].

Step 2 is to engage that community to develop metadata standards that signal two types of data quality: both how the data is produced and for what ends it can be used.

Data quality has multiple dimensions, reflecting both its production, which can be documented by the community of data producers, and its use, which can be documented by the community of users for different purposes. For example, data series may be perfectly valuable for some uses, but their combination and comparisons will result in highly erroneous conclusions. Metadata standards should make fitness-to-purpose clear to all potential users.

Much work has been done to develop metadata about how data are produced. NIH and other agencies have invested heavily in developing measures of how data are produced and providing data access through federally supported data repositories (e.g., the Generalist Repository Ecosystem Initiative[13]). Tools like Googe Dataset Search have also been developed to find datasets[14].

[10] Watson, Clare. "Many researchers say they'll share data—but don't." *Nature* [redacted] (2022): 853. https://www.nature.com/articles/d41586-022-01692-1

[11] Meng, Xiao-Li. "Data Democratization: An Ecosystemic Contemplation and Coordination." *Harvard Data Science Review* Special Issue 4 (2024). https://hdsr.mitpress.mit.edu/pub/o3qm160v/release/6?readingCollection=a6db7dff

[12] Lane, Julia, Alfred Spector, and Michael Stebbins. "An invisible hand for creating public value from data." *Harvard Data Science Review* Special Issue 4 (2024). https://hdsr.mitpress.mit.edu/pub/6blzwgpg/release/4?readingCollection=a6db7dff

[13] Generalist Repository Ecosystem Initiative

[14] Sostek, Katrina, et al. "Discovering datasets on the web scale: Challenges and recommendations for Google Dataset Search." *Harvard Data Science Review* Special Issue 4 (2024). https://hdsr.mitpress.mit.edu/pub/psnc8zsr?readingCollection=a6db7dff

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Recommendation: The Plan should recommend that agencies develop metadata about data use and reward their respective communities for their contribution. The Plan should support agencies in the development a community-driven contextual metadata framework that identifies domain-specific metadata requirements and contextual standards about data use and quality through community engagement. It should require the definition of quality indicators by use relevant to specific use cases, and that could be aligned with existing production standards, such as the FAIR principles[15] and the Federal Committee on Statistical Methodology framework[16]. It should also standardize a dataset annotation process, whereby annotation workflows could be implemented for pilot datasets in a scalable manner. The Plan should institutionalize validation procedures with expert communities, as well as the establishment of quality control processes, and feedback mechanisms to ensure metadata quality. The annotation patterns and best practices would need to be documented to foster scalability.

Step 3 is to create technology and incentives so that the community could contribute to developing information about data quality, so that data quality will rapidly improve and data usage becomes more informed and rigorous. This approach could mirror the approach developed by user platforms in the private sector, like Yelp, Airbnb, and Uber.

The result could be that the non-expert scientific research community and the public at large can have greater trust in the results. transform how users find and engage in research communities with common research questions and interests and enable them to explore data-driven approaches and methodologies based on the use of data. In the long term, the approach could be used to identify challenges in standardizing the ingestion of data extracted from different corpora into a unified dataset usage statistics schema and determine the future automated development and validation of these statistics.

Recommendation: The Plan could include investments in such technologies as the National Data Platform[17] and require that trust measures based on use be discovered, accessed and updated through federal funded and supported data repositories,

Please do not hesitate to contact me should you have any additional questions.

Sincerely,

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[15] https://www.go-fair.org/fair-principles/
[16] https://www.fcsm.gov/assets/files/docs/2023-conference-docs/F1.3_Mannshardt.pdf
[17] https://nationaldataplatform.org/